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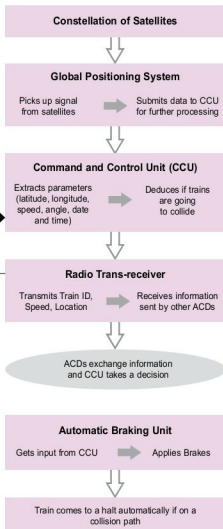


Fig. 3: Flowchart depicting the working of a GPS based ACD

nently along with the automatic braking unit.

Guard ACD (portable and moving). It is a portable ACD that acts as a backup to the locomotive ACD.

Station ACD (fixed). It is installed in the station master's office. He or she can send a manual SOS by pressing SOS buttons on the ACD on noticing something unusual.

Level-crossing gate ACD (fixed). It is installed in a gateman's cubicle and is used to inform other ACDs about any emergency at the level crossing.

Different scenarios for ACD application

Gate approach. Train approaches gate ST346. At approximately two kilometres from the gate, the message 'Gate Approach' flashes and

ST346 GATE APPROACH
19:26

Fig. 4: Driver console for gate approach scenario

ST346 GATE OPEN
19:26

Fig. 5: Driver console for gate open scenario

ST346 GATE OCCUPIED
19:26

Fig. 6: Driver console for gate occupied scenario

Advantages of ACD

- Acts automatically to prevent collisions
- System can withstand harsh environments
- Economical and cost-effective
- Elimination of human error as no inputs are to be fed by the crew

the audio buzzer sounds for ten seconds on the driver console.

Gate open. Level-crossing gate ST346 is detected to be open. Audio-visual indication is automatically activated to draw the attention of the driver, and brakes are applied to reduce the speed to 15 kilometres per hour.

Gate occupied. Level-crossing gate ST346 is detected to be occupied by a road vehicle. Audio-visual indication is automatically activated to draw the attention of the driver, and brakes are applied to reduce the speed to 15 kilometres per hour.

However, there are situations where ACD cannot prevent collision. Some examples are:

1. When one of the trains is non-ACD
2. Adequate braking distance is not available when collision situation arises, but impact of collision would be considerably reduced
3. Failure of brakes in trains

With the success of ACDs for trains, development of ACDs for cars is being considered to minimise road accidents. **EFY**